Design of an RF six-mode CMOS power amplifier for efficiency improvement at power backoff

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Abstract—**In this paper, the design and simulation results of a six-mode 130 nm CMOS power amplifier (PA) for communication links in 2.45 GHz are presented. The proposed two-stage PA is composed by independently programmable gain and power stages. Each stage is composed by three parallel amplifying cascode cells which are controlled by a digital word. At power backoff, when high output power levels are not required, a number of the parallel cells can be disabled allowing to improve the circuit efficiency. The designed layout has a 2 mm^2 die area. Post-layout simulation results show that not all the 64 possible operation modes should be used. Thus, a selection methodology was applied resulting on six operation modes. Among the six selected modes, the lowest power mode presents an 11.9 dBm output 1-dB compression point (OCP_{1dB}), a 22.2 dB small-signal power gain and a 214 mW power consumption. The highest power mode presents a 16.9 dBm OCP_{1dB}, a 31 dB gain and a 394 mW consumption. Therefore, for a 5 dB backoff from the highest OCP_{1dB}, the efficiency can be improved by 84% by using the low power mode.**

Keywords—2.45 GHz PA, 130 nm CMOS, Programmable PA, Variable gain, Variable power.

I. Introduction

Connected mobile devices are largely used nowadays, so that their number in 2018 is larger than the world population according to [1]. One of the main characteristics of most mobile devices is the wireless communication and the power supply through a battery. However, a critical point of these batteries is their lifetime, which can be improved by decreasing the device DC power consumption (P_{DC}).

Due to its large P_{DC}, the power amplifier (PA) is the most important active component of a wireless transmission chain. In [2], for instance, where Bluetooth is targeted, the impact of total power consumption of the PA in the chain varies between 36.2 % and 44.7 %. Additionally, this figure of merit may be as high as 67%, as [3] presents. In this sense, a design oriented to energy efficiency is a must and a leverage in such a populated market.

As linear PAs normally maintain a nearly constant P_{DC} for a wide range of output powers, overall efficiency decreases as output power is backed off. Different methods on achieving smart usage of energy are available in the literature, such as those presented in [4], [5] and [6].

In [4] a two-stage six-mode CMOS PA at 2.4 GHz with a fixed power stage and a variable gain stage is presented. The variable gain stage is composed of three parallel gain cells which may be enabled or disabled depending on the input power required. In the low-gain mode, this PA presented a 22 dB small-signal gain while consuming 96 mW at a 13 dBm output 1 dB refereed compression point (OCP_{1dB}). In the high-gain mode, it presented a 31 dB gain while consuming 191 mW P_{DC} at an OCP_{1dB} of 14 dBm.

In its turn, [5] proposed a multimode CMOS PA at 2.4 GHz, with a fixed gain stage and a single ended reconfigurable power stage. The power stage is composed of three parallel cascode power cells which can be enabled or disabled digitally by asserting high or low logic voltage in the common gate transistor. This PA achieves 13 dB gain and 6 dBm OCP_{1dB} at the lowest power mode while consuming 171 mW. For the highest power mode, a gain of 21 dB and an 18 dBm OCP_{1dB} are achieved while consuming 415 mW.

In [6] a two-stage four-mode CMOS PA at 2.5 GHz is presented. The power stage is based on four differential cascode power cells that may be enabled or disabled depending on the input signal envelope. The lowest power mode presented a 385 mW P_{DC} consumption and a 17 dBm OCP_{1dB} while the highest presented 2840 mW P_{DC} consumption and a 26 dBm OCP_{1dB}.

This paper presents the design of a two-stage reconfigurable six-mode 2.45 GHz CMOS PA. The topology of this PA is based on the reconfigurability in both gain and power stages. Such reconfigurability is achieved by employing singled ended cascode gain and power cells that are differently sized and enabled independently depending on the aimed output power. Section II presents the proposed topology and layout. In section III the methodology to select the power modes is presented. Section IV discusses the results of PA post-layout simulations and compare them to the reviewed literature. Lastly, Section V presents the main conclusions of this work.

II. Circuit Design

Usual PA architectures are composed by a gain stage, a power stage and impedance matching networks. Fig. 1 presents

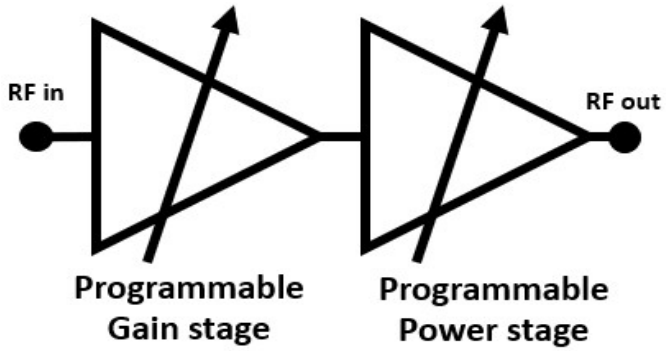

Fig. 1. Proposed circuit representation

the architecture of the proposed PA, where two programmable stages are shown, considering the impedance matching networks as parts of the stages. The first one (from [4]) is responsible mostly for the gain of the PA and the stage (from [5]) responsible for the output power.

The schematic of the proposed PA is represented in Fig. 2. The supply of the circuit is given by a 1.8 V voltage. The two programmable stages are composed by three cascode cells each. All the three cells of the gain stage are biased by the same voltage, configuring the cells in class AB. The same happens with the three cells in the power stage. The differences between the cells in both stages are the transistor widths, which provide distinct amplifying characteristics for each cell. Thus, the PA may have a total of 64 modes of operation with different gain, OCP_{1dB} and P_{DC} consumption.

The activation of the cells is realized through the control terminals, applying VDD into the terminal to activate the cell or applying 0 V to deactivate it. Since the control signals are binary, a logic circuit could be implemented to control the switching between the different operation modes.

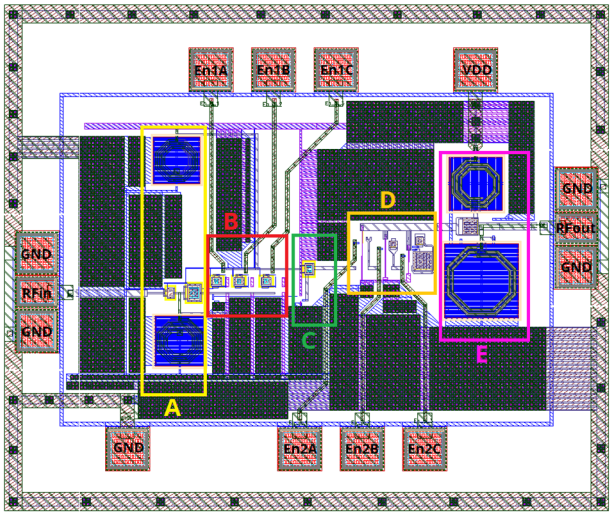

Fig. 3. Proposed PA layout. (A): Input impedance matching network. (B): Programmable gain stage. (C): Interstage impedance matching capacitor. (D): Programmable power stage. (E): Output impedance matching network.

The layout for the proposed circuit is represented in Fig. 3. The die area is 1.7 mm x 1.2 mm, including RF pads and DC pads for VDD and GND and for activation of the cascode cells (En1A, En2A, En3A, En1B, En2B and En3B). For the adopted technology (Global Foundries RF 130 nm CMOS), eight layers of metals are available to perform routing. Ground and supply planes were placed in and around the circuit utilizing all metal layers to minimize parasitic losses. ESD protection diodes were positioned close to control pads. The transistors gates are segmented in several fingers, connected in both sides to reduce the gate resistance. In addition, the active area of the PA was laid out in a compact way to reduce access resistances.

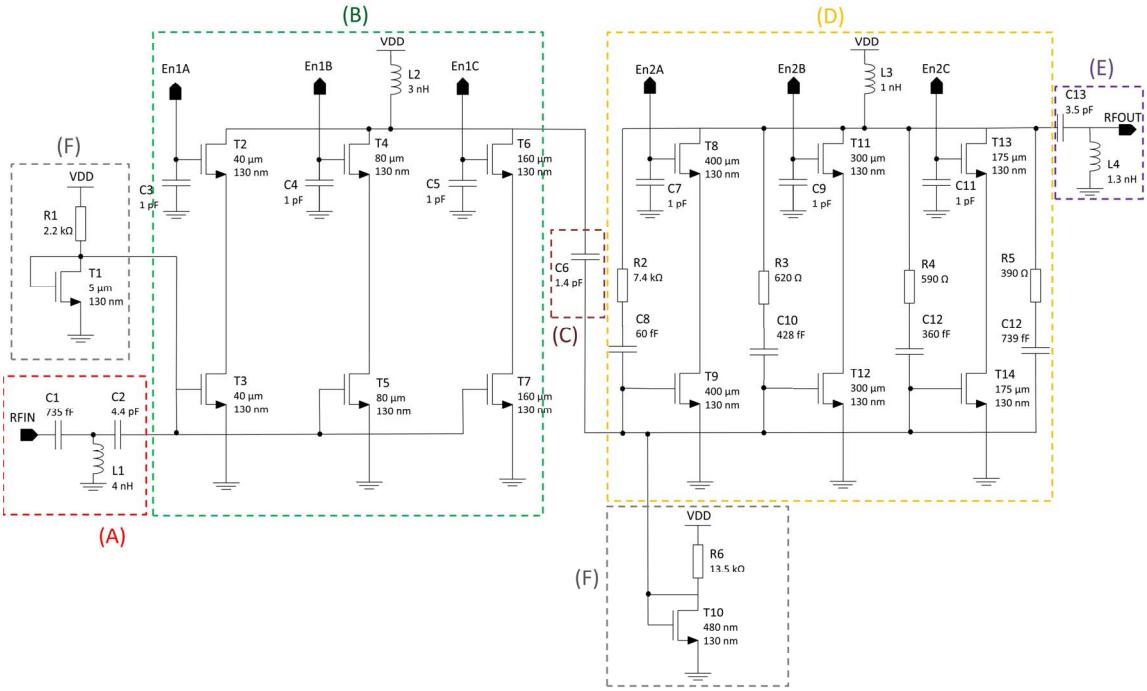

Fig. 2. Proposed PA schematic. (A): Input impedance matching network.. (B): Programmable gain stage. (C): Interstage impedance matching. (D): Programmable power stage. (E): Output impedance matching. (F): Bias circuitry.

III. Mode Selection

Both programmable stages used in the proposed PA present 8 possible operation modes. Thus, the proposed circuit presents a total of 64 possible combinations. However, there are some modes that should be discarded because their characteristics do not justify their use, so a selection methodology was applied to determine the modes of interest. This methodology is summarized in the flow chart on Fig. 4.

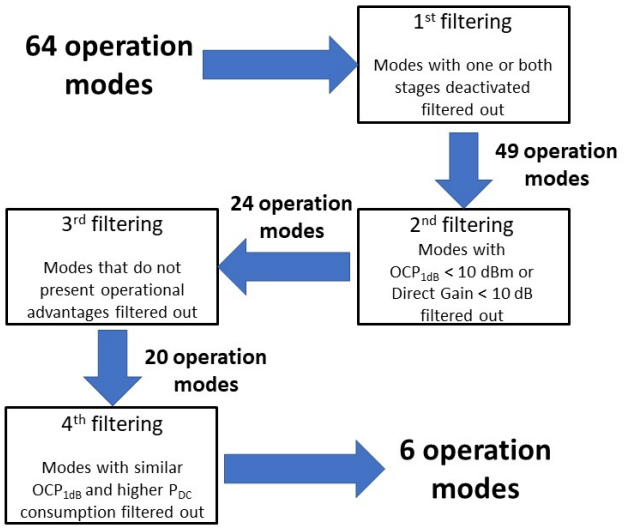

Fig. 4. Mode selection methodology flow chart

Post-layout simulations were carried out using 50 Ω ports in the input and output terminals. Scattering parameters and harmonic balance simulations were performed, both using Cadence Spectre RF.

The first criterion applied was to exclude all the modes for which one or both stages were totally deactivated. The remaining 49 modes are represented in Table I. The second filtering consisted in discarding the modes with OCP_{1dB} below 10 dBm or a direct gain lesser than 10 dB. After the second filter was applied, 24 operating modes remained.

Although all the remaining modes presented reasonably good characteristics, when compared to other operation modes some of them did not present operational advantages. An example of a mode that does not present operational advantages and should be discarded can be observed in Table I. Mode 100100 presents lower values of direct gain and OCP_{1dB} than the mode 010011, but it has a greater P_{DC} consumption. This means that the 100100 mode consumes more energy to amplify less than 010011 mode and to deliver less power. Filtering out the modes that do not present operational advantages was the third filter applied.

The 20 remaining modes were further filtered in order to simplify the future implementation of a logic control of the modes. The fourth filtering consists in discarding modes with similar OCP_{1dB} values, with a difference lesser than 1 dB, that present a higher P_{DC} consumption. An example of this filtering can be observed comparing the modes 010011 and 111011. The 010011 mode presents a similar OCP_{1dB} and lower power consumption than 111011 so, the latter was filtered out. The six

TABLE I. REMAINING OPERATION MODES AFTER THE FIRST FILTERING

Operation mode						Metrics		
En1A	En1B	En1C	En2A	En2B	En2C	OCP1 (dBm)	Gain (dB)	PDC (mW)
0	0	1	0	0	1	4.7	20.2	139
0	0	1	0	1	0	8.1	23.2	166
0	0	1	0	1	1	12.4	27.3	227
0	0	1	1	0	0	13.3	28.	249
0	0	1	1	0	1	15.3	30.1	310
0	0	1	1	1	0	15.9	30.9	337
0	0	1	1	1	1	16.7	32.2	398
0	1	0	0	0	1	4.1	15.0	126
0	1	0	0	1	0	7.5	18.	153
0	1	0	0	1	1	11.9	22.2	214
0	1	0	1	0	0	12.9	22.9	236
0	1	0	1	0	1	15.	25.	297
0	1	0	1	1	0	15.7	25.9	324
0	1	0	1	1	1	16.6	27.1	385
0	1	1	0	0	1	4.6	24.6	154
0	1	1	0	1	0	8.0	27.2	181
0	1	1	0	1	1	12.4	31.2	242
0	1	1	1	0	0	13.0	31.9	265
0	1	1	1	0	1	14.9	33.9	326
0	1	1	1	1	0	15.4	34.7	353
0	1	1	1	1	1	16.2	36.0	414
1	0	0	0	0	1	2.1	10.1	119
1	0	0	0	1	0	5.2	13.2	146
1	0	0	0	1	1	9.5	17.2	207
1	0	0	1	0	0	10.3	18.0	229
1	0	0	1	0	1	12.4	20.1	291
1	0	0	1	1	0	13.2	20.9	318
1	0	0	1	1	1	14.3	22.3	378
1	0	1	0	0	1	4.6	22.7	147
1	0	1	0	1	0	8.0	25.7	175
1	0	1	0	1	1	12.3	29.7	236
1	0	1	1	0	0	13.2	30.4	258
1	0	1	1	0	1	15.1	32.4	319
1	0	1	1	1	0	15.7	33.3	346
1	0	1	1	1	1	16.5	34.5	407
1	1	0	0	0	1	4.5	19.0	134
1	1	0	0	1	0	7.9	22.0	162
1	1	0	0	1	1	12.3	26.1	223
1	1	0	1	0	0	13.3	26.8	245
1	1	0	1	0	1	15.4	28.9	306
1	1	0	1	1	0	16.0	29.7	333
1	1	0	1	1	1	16.9	31.	394
1	1	1	0	0	1	4.3	25.9	163
1	1	1	0	1	0	7.6	28.9	190
1	1	1	0	1	1	11.7	32.8	251
1	1	1	1	0	0	12.6	33.5	274
1	1	1	1	0	1	14.4	35.5	335
1	1	1	1	1	0	15.0	36.3	362
1	1	1	1	1	1	15.8	37.5	422

TABLE II. SIX SELECTED MODES AFTER FILTERING

Operation mode							Metrics		
Name	En1A	En1B	En1C	En2A	En2B	En2C	OCP1 (dBm)	Gain (dB)	PDC (mW)
A	0	1	0	0	1	1	11.9	22.2	214
B	0	1	0	1	0	0	12.9	22.9	236
C	1	1	0	1	0	0	13.3	26.8	245
D	0	1	0	1	0	1	15.	25.	297
E	0	1	0	1	1	0	15.7	25.9	324
F	1	1	0	1	1	1	16.9	31.	394

selected modes were named from "A" to "F" as presented in Table II.

It also can be observed in the Table II, that the control signal En1B has always the value 1 and the control signal En1C has always the value 0. Thus, the gain cell controlled by En1C can be obviated and the En1B signal can be excluded from the control signals.

IV. RESULTS AND DISCUSSION

Detailed post-layout simulations were carried out for the 6 selected operating modes using S-parameter and harmonic balance in Cadence Spectre RF.

Fig. 5 presents the stability factor (μ) of the six selected operation modes. The proposed circuit is unconditionally stable for all modes because the value of μ is never lower than one. The graph is clipped to a maximum μ value of 10 for a better visualization of the region closer to μ = 1. The peak value of the stability factor obtained in the simulation was 40, for the operation mode F.

Fig. 6 presents the direct small-signal gain, given by the scattering parameter S_{21}, of the six selected operation modes. In detail, the gain at 2.45 GHz is presented. The gain can be varied from 22.2 dB to 31 dB, representing an 8.8 dB range. It also can be seen that the direct gain is slightly higher in 2.5 GHz and slightly lower in 2.4 GHz when compared to the value in 2.45 GHz. This occurs for all selected operation modes. For example, mode F presents a 30.7 dB direct gain at 2.4 GHz, 31 dB at 2.45 GHz and 31.2 dB at 2.5 GHz.

Fig. 7 presents P_{OUT} as a function of the input power (P_{IN}) for all the selected modes. It can be observed that OCP_{1dB} values vary from 11.9 dBm to 16.9 dBm (5 dB range). P_{DC} of the six selected modes vary from 214 mW to 394 mW, as presented in Fig. 8. This variation in the power consumption can result in a

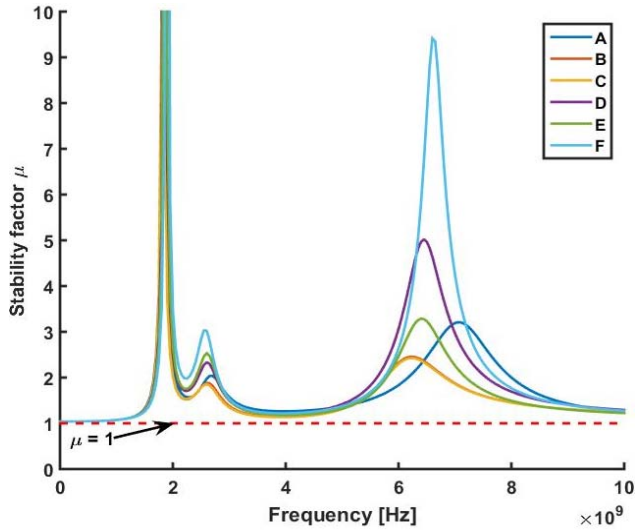

Fig. 5. Stability factor (μ) of the six operating modes of the PA.

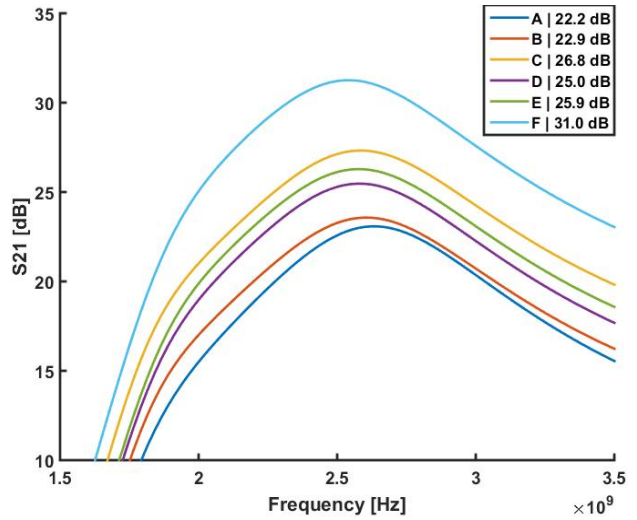

Fig. 6. Small-signal direct gain vs. frequency for the six operating modes of the PA.

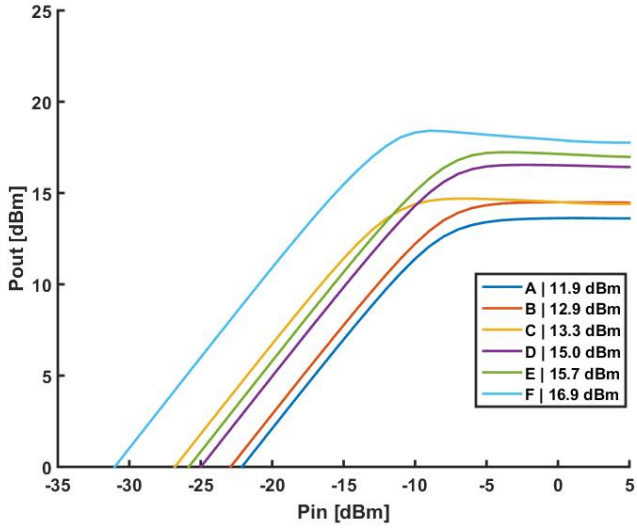

Fig. 7. P_{OUT} vs. P_{IN} at 2.45 GHz for the six operating modes of the PA, with OCP_{1dB} values.

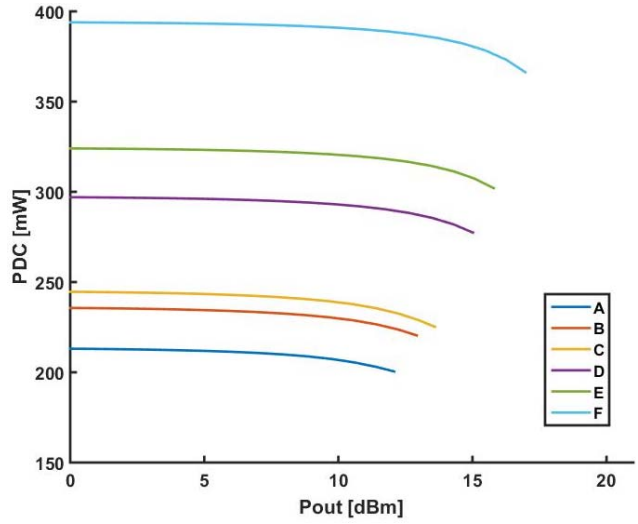

Fig. 8. P_{DC} vs. P_{OUT}. The curves are clipped at OCP_{1dB} for each mode.

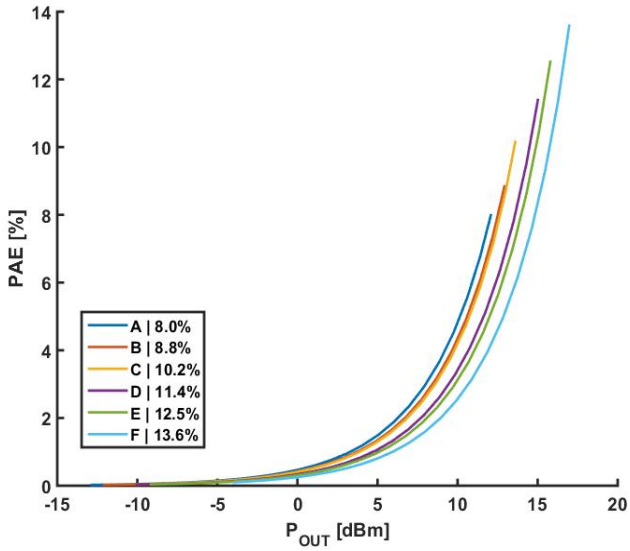

Fig. 9. PAE vs.P_{out} for the six operating modes of the PA and their values at OCP_{1dB}.

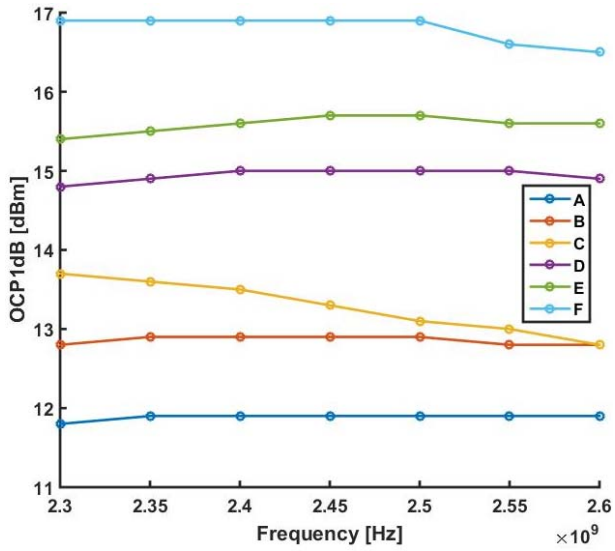

Fig. 10. OCP_{1dB} vs. frequency for the six operating modes of the PA.

power saving on the RF module, one of the most important characteristics of the proposed circuit.

For example, if linear operation is required for a 2 dB backoff from the peak OCP_{1dB}, i.e. at 14.9 dBm, modes D, E and F are eligible. Modes A, B and C are not capable of achieving such P_{OUT}. Among the possible modes, mode D would be the selected operation mode, as this mode can deliver the aimed P_{OUT} while consuming the least P_{DC} (297 mW), if compared to modes E (324 mW) and F (394 mW). If mode D is compared to mode F, approximately 100 mW are saved.

Fig. 9 presents the simulated PAE vs. P_{OUT} for the six selected modes up to their individual OCP_{1dB}. For instance, for a 15 dBm P_{OUT}, the most efficient mode would be D with a PAE of 11.4%. This figure also presents the potential improvement of efficiency as P_{OUT} is backed off and operation can be switched to a more suitable mode.

Fig. 10 shows the values of OCP_{1dB} in function of the PA operation frequency for all modes. The OCP_{1dB} values, in general, present very small variations between 2.3 GHz and 2.6 GHz. Mode C is the exception, with OCP_{1dB} ranging from 13.7 dBm to 12 dBm. Fig. 11. depicts the drain efficiency variation at OCP_{1dB} over frequency, for the six operating modes. The graph behavior is similar to the graph of OCP_{1dB} in function of the frequency. Except for mode C, the variations in the 2.4 GHz – 2.5 GHz range are slight.

Table III presents a comparison to similar works in the literature. Since [4] only reconfigures the gain stage, mode switching does not allow very significant power savings or variations on the linear output power. This is the case for the output power in [7] as well, which is highest among the presented papers, but cannot be significantly reduced. For this reason, even for the low-gain mode, power consumption is higher than in our work. The work in [5] presents similar results to the proposed circuit, but with a significantly lower gain, and [9] also presents a good combination of performance metrics, but with lower power gain values than this work. [6] presents very

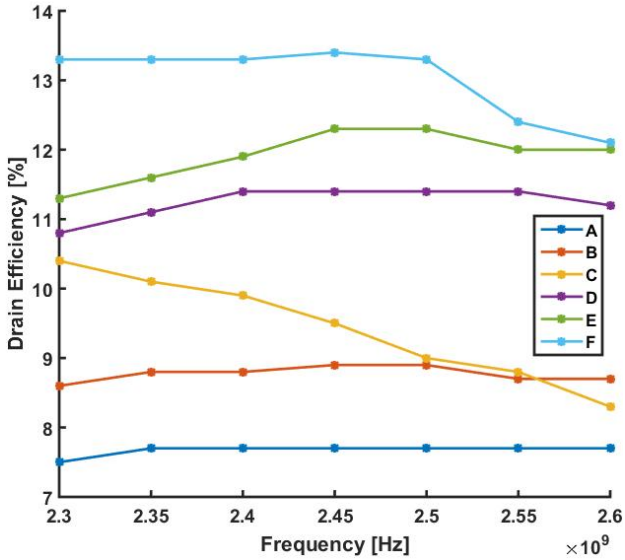

Fig. 11. Drain efficiency at OCP_{1dB} vs. frequency for the six operating modes of the PA.

good results in terms of output power and gain, but this comes at the cost of a significantly high power consumption and a much larger area, due to the use of a bulky transformer-based power combiner. The circuit presented in [10] is another example of a realization which achieves more than 20 dBm at OCP_{1dB} in the high power mode but with a larger area occupation, larger power consumption and lower PAE. Finally, the work in [8] presents the lowest P_{DC} values, the highest PAE and excellent performance in all considered aspects. Its drawback, though, is that reconfiguration is limited to two operating modes, which restricts its application at intermediate conditions. Therefore, the PA proposed in this work represents an interesting trade-off in terms of gain, power consumption, linearity and number of operating modes.

TABLE III COMPARISON WITH STATE OF THE ART

Ref	Tech. node (nm)	Frequency (GHz)	Number of modes	Area (mm^2)	Gain (dB)	P_{DC} (mW)	OCP_{1dB} (dBm)	PAE @ OCP_{1dB} (%)
[4]	130	2.4	6	1.2	22 - 31	171 – 196	14	13 – 15[2]
[5][1]	130	2.4	7	1.7	13 - 21	171 - 415	6 - 18	2 - 17
[6]	65	2.4	4	4.9	16 – 24[2]	385 – 2840	17 - 26	5 – 12[2]
[7]	180	2.4	2	2	20 - 38	429 – 2574	31	10 – 27[2]
[8]	180	2.4	2	0.88	15 - 25	40 – 132	16 - 22	28 - 40
[9]	350	0.915	7	0.5	11 – 27	186 – 236	9.3 – 20.3	4 - 37
[10]	90	5.5	4	2.4	17 - 26[2]	360 – 1800[2]	9.5 – 21.7	2 - 8[2]
This work[1]	**130**	**2.4**	**6**	**2**	**22 - 31**	**214 – 394**	**12 - 17**	**8 - 14**

1- Simulated results, 2 - Estimated

V. CONCLUSION

In this paper the post-layout simulation results of a fully integrated PA with a gain and a power programmable stages were presented. The combination of these stages may provide a total of 64 different combinations by enabling or disabling parallel cascode cells. However not all the possible modes justify their use, so that the six modes which present most advantages (higher OCP_{1dB} and direct gain values and less P_{DC} consumption) were selected with a four-step selection. The six selected modes present a direct gain from 22.2 dB to 31.0 dB, OCP_{1dB} from 11.9 dBm to 16.9 dBm and P_{DC} varying from 214 mW to 394 mW. The utilization of the low power mode can result into an energy economy up to 45.7% when compared with the high power mode in backoff operation. This economy on the PA module can result on significant power consumption reduction on the RF transceivers and, consequently, the improvement of the mobile device battery lifetime. Therefore, in comparison to state-of-the-art realizations, the proposed PA architecture with two programmable stages, presents a good alternative to design efficient multimode PAs.

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